

MISKAYE HIZUNAN MEDHANEALEM MONASTERY SCHOOL
2018 E.C. CHEMISTRY MODEL EXAM FOR GRADE 12 STUDENTS

Time allowed:- 90 min.

PHYSICAL CONSTANTS:-

- Gas constant, $R=8.314\text{J/molK}=0.0821\text{L.atm/molK}$
- Avogadro's number = $6.023 \times 10^{23}\text{mol}^{-1}$
- Planck's constant, $h=6.626 \times 10^{-34}\text{JS}$
- Speed of light, $C=3 \times 10^8\text{ m/s}$
- Faraday's constant(F) = 96500C/mol

ATOMIC NUMBERS (Z) AND ATOMIC WEIGHTS (A)

Element	H	C	N	O	F	Na	Mg	Al	P	S	Cl	K	Ca	Mn	Fe	Ni	Cu
Z	1	6	7	8	9	11	12	13	15	16	17	19	20	25	26	28	29
A	1	12	14	16	19	23	24	27	31	32	35.5	39	40	55	56	59	65.5

Direction:- This exam paper consist 80 multiple choice questions. Attempt to read all questions carefully then select the correct answer among the given alternative and put the Choice of your answer on the separate answer sheet provided (1Pt each).

1. Who proposed the Law of Octaves, stating that every eighth element had similar properties with the first element?
A. Dmitri Mendeleev B. John Newlands C. J. Dobereiner D. Henry Moseley
2. Which of the following was the major contribution of Moseley to the modern periodic table?
A. He discovered the atomic mass of elements
B. He established that elements should be arranged by atomic number, not atomic mass
C. He proposed the Law of Octaves
D. He arranged elements into triads
3. Which of the following statements is true about periods in the modern periodic table?
A. All periods contain 8 elements
B. Period 7 is complete
C. Elements in the same period have the same number of shells
D. Periods 1 and 2 are known as long periods
4. What was a key prediction made by Mendeleev regarding undiscovered elements?
A. He left spaces for undiscovered elements and predicted their properties
B. He predicted their exact atomic mass
C. He predicted their atomic number
D. He predicted their position based on isotopes
5. Why does the first period of the periodic table contain only two elements?
A. The first period only has two groups of elements.
B. The first period consists only of metals.
C. The first period fills only the 1s orbital, which can hold a maximum of two electrons.
D. The first period is incomplete due to missing elements.
6. What is the electron configuration of chlorine (Cl), and how many valence electrons does it have?
A. $1s^2 2s^2 2p^6 3s^2 3p^6$, 7 valence electrons C. $1s^2 2s^2 2p^6 3s^2 3p^4$, 6 valence electrons
B. $1s^2 2s^2 2p^6 3s^2 3p^6$, 5 valence electrons D. $1s^2 2s^2 2p^6 3s^2 3p^5$, 7 valence electrons
7. What is the general trend in atomic size as you move down a group in the periodic table?
A. Atomic size decreases because the number of protons increases.
B. Atomic size increases because the number of protons decreases.
C. Atomic size increases because the number of electron shells increases.
D. Atomic size remains constant as you move down a group.
8. Which of the following elements has the highest electron affinity?
A. Sodium (Na) B. Fluorine (F) C. Chlorine (Cl) D. Oxygen (O)
9. How does the periodic table help predict the behavior of compounds?
A. It tells us the exact price of elements used in compounds.
B. It shows how oxides of elements become more acidic across a period and more basic down a group.

- C. It predicts the exact volume of gases produced in reactions.
 D. It calculates the exact time required for compounds to react.
10. Why do halogens have the highest electron affinities among the elements?
 A. They have a low atomic size and high nuclear charge.
 B. They easily lose electrons to form positive ions.
 C. They have a half-filled or completely filled outer shell.
 D. They tend to form positive ions with a strong attraction to electrons.
11. Which of the following compounds is classified as an acidic oxide?
 A. Sodium oxide (Na_2O) C. Sulphur dioxide (SO_2)
 B. Magnesium oxide (MgO) D. Calcium oxide (CaO)
12. What is the primary difference between the Brønsted-Lowry and Arrhenius definitions of acids?
 A. Brønsted-Lowry acids release H^+ ions in aqueous solutions.
 B. Brønsted-Lowry acids are proton donors, while Arrhenius acids focus on H^+ in aqueous solutions.
 C. Arrhenius acids can accept electron pairs.
 D. Brønsted-Lowry acids only exist in aqueous solutions.
13. What will happen when an acid reacts with a carbonate or hydrogen carbonates? It can form
 A. It forms salt and hydrogen gas and carbon monoxide.
 B. It forms salt, water, and carbon dioxide gas.
 C. It forms water and hydrogen gas only.
 D. It forms salt and oxygen gas.
14. What is the key difference between a concentrated acid and a dilute acid?
 A. A concentrated acid has more water than acid, whereas a dilute acid has more acid than water.
 B. A concentrated acid contains a larger amount of acid in a smaller volume of solution than a dilute acid.
 C. A concentrated acid contains more ions than a dilute acid.
 D. A dilute acid is stronger than a concentrated acid.
15. When diluting concentrated acids, which of the following is the correct procedure?
 A. Pour the acid into water. C. Mix acid and water in equal amounts.
 B. Pour water into the acid. D. Mix acid with a non-volatile acid.
16. If two solutions, A and B, have pH values of 2 and 6 respectively, how many times greater is the hydrogen ion concentration in solution A than in solution B?
 A. 10 B. 10000 C. 100 D. 1000
17. Which of the following acids is used in the manufacture of inorganic fertilizers?
 A. Hydrochloric acid B. Nitric acid C. Sulfuric acid D. Acetic acid
18. Which industrial process uses sulfuric acid (H_2SO_4) as an electrolyte?
 A. Petroleum refining C. Production of metals
 B. Electroplating of metals D. Car batteries
- 4
19. If the concentration of OH^- is $1.0 \times 10^{-11} \text{ M}$ in an aqueous solution at 25°C , what is the concentration of H_3O^+ and pH?
 A. $1.0 \times 10^{-11} \text{ M}$, 3 C. $2.0 \times 10^{-11} \text{ M}$, 4
 B. $1.0 \times 10^{-10} \text{ M}$, 10 D. $1.0 \times 10^{-3} \text{ M}$, 11
20. Which of the following is a characteristic property of bases?
 A. Bases turn red litmus paper blue.
 B. Bases are acidic to the touch.
 C. Bases release hydrogen ions (H^+) in aqueous solutions.
 D. Bases have a sour taste.
21. Which of the following is the most important immediate action to take if a base is splashed into the eyes during a laboratory experiment?
 A. Apply a dilute solution of acetic acid to neutralize the base.
 B. Rinse the eyes with water continuously for at least 15 minutes and seek medical attention.
 C. Immediately cover the eyes with a sterile gauze pad and seek medical attention.
 D. Take a break and allow the eyes to rest until the irritation subsides.

22. Which of the following is NOT a correct use or characteristic of calcium hydroxide (Ca(OH)_2) according to the text?
- A. It is used to make mortar, which binds bricks and is used in plastering walls.
 - B. It is a stronger base than sodium hydroxide and is used in organic reactions.
 - C. It is used to remove temporary hardness of water and in manufacturing bleaching powder.
 - D. Its aqueous solution is used to test for the presence of carbon dioxide gas.
23. Which of the following an example of an acidic salt?
- A. Sodium chloride (NaCl)
 - B. Sodium sulfate (Na_2SO_4)
 - C. Sodium hydrogen sulfate (NaHSO_4)
 - D. Zinc hydroxychloride (Zn(OH)Cl)
24. When an active metal like magnesium reacts with dilute hydrochloric acid, which of the following products is not formed in the reaction?
- A. Salt (MgCl_2)
 - B. BHydrogen gas (H_2)
 - C. Water (H_2O)
 - D. The reaction produces no other product besides magnesium chloride.
25. Which of the following statements about deliquescent salts is correct?
- A. Deliquescent salts absorb water from the atmosphere but remain solid.
 - B. Deliquescent salts absorb water from the atmosphere and form a solution.
 - C. Deliquescent salts lose their water of crystallization to the atmosphere.
 - D. All hygroscopic salts are also deliquescent.
26. Which of the following statements about the role of nitrogen in plant growth is incorrect?
- A. Nitrogen is essential for synthesizing amino acids, proteins, and nucleic acids in plants.
 - B. Nitrogen is absorbed by plants primarily in the form of nitrate ions (NO_3^-).
 - C. Nitrogen plays a role in the synthesis of chlorophyll and contributes to a deep green color in plants.
 - D. Nitrogen promotes the growth of roots and facilitates early seed production.
27. Which of the following inorganic pesticides is primarily used as a fungicide, according to the text?
- A. Hydrogen cyanide (HCN)
 - B. Copper sulfate (CuSO_4)
 - C. Ammonium sulphamate ($(\text{NH}_4)\text{SO}_3\text{NH}_2$)
 - D. Sodium chlorate (NaClO_3)
28. Which experiment led Rutherford to propose that an atom has a dense, positively charged nucleus?
- A. Cathode ray experiment
 - B. Gold foil experiment.
 - C. Mass spectrometer experiment.
 - D. Oil drop experiment.
29. Which of the following statements best describes the trend in metallic character across a period?
- A. Metallic character increases as you move from left to right across a period.
 - B. Metallic character decreases as you move from left to right across a period
 - C. Metallic character does not change across a period.
 - D. Metallic character is highest in the noble gases.
30. What is the primary difference between alpha (α), beta (β), and gamma (γ) rays in terms of their properties?
- A. Alpha particles have no charge, beta particles are positively charged, and gamma rays are electrons.
 - B. Alpha particles are positively charged, beta particles are negatively charged, and gamma rays are neutral and not affected by electric or magnetic fields.
 - C. Alpha particles are neutral, beta particles are positively charged, and gamma rays have a negative charge.
 - D. Alpha rays are unaffected by electric and magnetic fields, whereas beta rays are deflected by both fields.
31. Which of the following compounds would have the lower lattice energy?
- A. NaF
 - B. MgO
 - C. LiCl
 - D. CaO
32. According to Planck's equation $E=h\nu$ if the frequency of a photon is doubled, what happens to the energy of the photon?
- A. The energy is halved
 - B. The energy is doubled
 - C. The energy remains the same
 - D. The energy quadruples
33. The energy from the Sun reaches the Earth mainly in the form of:
- A. Gamma rays and microwaves
 - B. Radio waves and X-rays
 - C. Visible and ultraviolet radiation
 - D. Infrared and gamma radiation

34. Which of the following correctly describes the allowed values of the angular momentum quantum number ℓ for a principal quantum number $n=3$?
- A. $\ell=0,1,2,3$ B. $\ell=1,2,3$ C. $\ell=0,1,2$ D. $\ell=0,1,2,3,4$
35. Why do metals tend to form cations, while non-metals tend to form anions?
- A. Metals lose electrons easily because they have low ionization energies, while non-metals gain electrons due to high electron affinities.
B. Metals lose electrons because they have high ionization energies, while non-metals gain electrons due to low electron affinities.
C. Metals tend to gain electrons, and non-metals lose electrons.
D. Metals and non-metals both tend to gain electrons
36. Which of the following statements best explains why the formation of a sigma (σ) bond requires a head-to-head overlap of atomic orbitals?
- A. The overlap must be perpendicular to the internuclear axis to maximize the electron density between the nuclei.
B. The s orbitals overlap side by side, leading to greater electron density along the bond axis.
C. A head-to-head overlap places the electron density along the bond axis, resulting in the strongest bond formation.
D. The p orbitals can only overlap side-to-side to form a sigma bond due to their shape.
37. For a molecule with polar covalent bonds, the overall dipole moment depends on the:
- A. Number of bonds in the molecule C. Ionization energy of the atoms
B. Geometry of the molecule D. Average bond length of the bonds
38. In sp hybridization, what percentage of s-character and p-character is present in each of the hybrid orbitals formed?
- A. 33% s-character, 67% p-character C. 67% s-character, 33% p-character
B. 75% s-character, 25% p-character D. 50% s-character, 50% p-character
39. What molecular shape does water (H_2O) adopt based on the VSEPR theory, considering it has two bonding pairs and two lone pairs?
- A. Tetrahedral C. Trigonal planar
B. Bent D. Trigonal pyramidal
40. What molecular geometry is predicted for a molecule with the formula AB_5 , where A is the central atom and there are no lone pairs on A?
- A. Trigonal planar C. Trigonal bipyramidal
B. Tetrahedral D. Octahedral
41. Which of the following is true about the hydrolysis of salts formed from strong acids and strong bases?
- A. The anions derived from strong acids undergo hydrolysis, resulting in an acidic solution.
B. The cations derived from strong bases undergo hydrolysis, resulting in a basic solution.
C. The solution formed from the reaction of a strong acid and a strong base will be neutral because neither the cation nor the anions undergo hydrolysis.
D. Both the anions and cations from strong acid-strong base salts undergo hydrolysis, producing a neutral solution.
42. How does the color of an acid-base indicator change with respect to the pH of the solution it is added to?
- A. The color of the indicator does not change unless a large amount is added, in which case the solution's pH is affected.
B. The color of the indicator depends on whether the solution is acidic or basic, with the indicator shifting to its non-ionized form in acidic conditions and to its ionized form in basic conditions.
C. The indicator always remains the same color regardless of the pH of the solution.
D. The color change of the indicator is due to a shift in the ionization of water, not the indicator itself.
43. Which of the following statements about Arrhenius bases is true?
- A. Arrhenius bases increase the concentration of H^+ ions in aqueous solution.
B. All metal hydroxides are Arrhenius bases.
C. Arrhenius bases cannot exist in non-aqueous solutions
D. Arrhenius bases completely dissociate to form OH^- ions in aqueous solution.

44. In the reaction: $\text{H}_2\text{PO}_4^- + \text{OH}^- \rightleftharpoons \text{HPO}_4^{2-} + \text{H}_2\text{O}$ which of the following is true?
- OH^- is the acid, and H_2O is the conjugate base.
 - H_2PO_4^- is an acid, and HPO_4^{2-} is its conjugate base.
 - H_2PO_4^- is a base, and HPO_4^{2-} is its conjugate acid.
 - H_2O is the acid, and OH^- is its conjugate base.
45. Which of the following is true about the strength of an acid and its acid dissociation constant (K_a)?
- A higher K_a value indicates a weaker acid.
 - A lower K_a value indicates a stronger acid.
 - A higher K_a value indicates a stronger acid.
 - K_a values do not affect the acid strength.
46. What is the effect of adding a base to a buffer solution containing acetic acid and acetate ions?
- The base will react with the acetic acid to form more acetate ions.
 - The pH will decrease.
 - The pH will increase.
 - The buffer will neutralize the base and the pH will stay constant.
47. What happens to the ionization of acetic acid when HCl is added to the solution?
- The ionization of acetic acid increases.
 - The ionization of acetic acid decreases.
 - There is no change in the ionization of acetic acid.
 - The concentration of acetate ions (CH_3COO^-) increases.
48. In a titration of a strong acid with a strong base, what would you expect the indicator's color change to be near the equivalence point?
- The color will change from red to blue.
 - The color will change from yellow to colorless.
 - The color will change from pink to colorless.
 - The color will change from red to yellow.
49. If 50.0 mL of 0.100 M NaOH is required to neutralize 25.0 mL of HCl, what is the molarity of the HCl solution?
- 0.200 M
 - 0.0500 M
 - 0.100 M
 - 0.250 M
50. Which of the following is the correct sequence of steps when using the change in oxidation-number method to balance redox reactions?
- Assign oxidation numbers, balance the charges, write an unbalanced chemical equation, and then identify the oxidized and reduced species.
 - Write an unbalanced equation, assign oxidation numbers, identify the oxidized and reduced species, compute the number of electrons, and balance the equation by inspection.
 - Write an unbalanced equation, compute the number of electrons lost and gained, identify the species, and then complete the equation by inspection.
 - Assign oxidation numbers, compute the electrons gained and lost, write an unbalanced equation, balance the charges, and then complete the equation by inspection.
51. Which factor primarily affects the preferential discharge of ions during electrolysis?
- The charge of the ion.
 - The size of the ion.
 - The temperature of the solution.
 - The position of the ion in the electrochemical series.
52. Which of the following factors primarily determines which ions get preferentially discharged at the electrodes during electrolysis?
- The type of electrode material used
 - The pH level of the solution
 - The concentration of ions in the solution
 - The temperature of the electrolyte solution
53. Two cells are connected in series. One contains AgNO_3 , and the other MF. If 1.44g of Ag was deposited in the cell containing AgNO_3 , whereas 0.12g of the unknown metal M was deposited in the cell containing MF_3 what will be the molar mass of M? (mass of silver = 108)
- 9
 - 27
 - 54
 - 3
54. A certain amount of electricity passes through a cell in which 30g Fe is deposited from FeSO_4 (aq). How much Au would be deposited if the same amount of electricity passed through $\text{K}[\text{AuCl}_4](\text{aq})$? (mass of iron = 56, mass of gold = 197)
- 27.9g
 - 32.8g
 - 70.4g
 - 98.5g

55. What is the main difference between a mineral and an ore?
- A. A mineral is any naturally occurring substance, whereas an ore is specifically a compound from which a metal can be extracted economically.
 - B. An ore always contains gold, while a mineral contains other metals such as aluminum.
 - C. A mineral is always in its pure form, while an ore is always in the combined form with other elements.
 - D. An ore is any compound that contains carbon, while a mineral contains only non-metallic elements.
56. What is the primary function of the external conducting medium (metal wire) in a galvanic cell like the Daniell cell?
- A. To provide a medium for ions to migrate between the two half-cells.
 - B. To allow the flow of electrons between the two electrodes, generating an electric current.
 - C. To separate the oxidizing and reducing agents to prevent the reaction from occurring.
 - D. To increase the concentration of ions in the solution to speed up the reaction.
57. Why the standard hydrogen electrode (SHE) is assigned a potential of exactly 0V?
- A. Because hydrogen is the most abundant element in the universe.
 - B. Because hydrogen ions in the solution are the easiest to reduce.
 - C. Because the platinum electrode does not participate in the reaction, thus no voltage is generated.
 - D. Because it serves as a reference electrode to compare the potentials of other electrodes under standard conditions.
58. Which of the following statements correctly describes the changes that occur in the concentration cell as it operates over time?
- A. The concentration of Cu^{2+} in both the anode and cathode compartments increases until equilibrium is reached.
 - B. The concentration of Cu^{2+} in the anode compartment decreases, while the concentration of Cu^{2+} in the cathode compartment increases.
 - C. The concentration of Cu^{2+} in the anode compartment increases, while the concentration in the cathode compartment decreases.
 - D. The concentration of Cu^{2+} in both compartments decreases equally, resulting in no change in the cell potential.
59. Which of the following correctly explains why the standard cell potential (E°_{cell}) for a concentration cell is zero?
- A. The concentrations of the Cu^{2+} ions in both half-cells are the same.
 - B. The concentration of Cu^{2+} ions in the anode compartment is equal to that in the cathode compartment.
 - C. The half-reactions are identical, with the same reduction potential for both the anode and cathode.
 - D. The half-reactions are different, so their electrode potentials cancel each other out.
60. What is the main role of natural resources in the chemical industry?
- A. They are refined to produce raw materials for further processing in various industries.
 - B. They are only used as energy sources.
 - C. They are used solely for food production.
 - D. They are only utilized in the pharmaceutical industry.
61. Which of the following is an example of a non-renewable resource?
- A. Solar energy
 - B. Wood
 - C. Coal
 - D. Fish
62. Which of the following is the industrial method used for the production of ammonia?
- A. Haber-Bosch process
 - B. Electrolysis of water
 - C. Fractional distillation of air
 - D. Reacting ammonium salts with strong alkalis
63. What is the main product of the Ostwald process?
- A. Urea
 - B. Ammonia
 - C. Nitrous oxide
 - D. Nitric acid
64. Why anhydrous ammonia is injected 6 to 8 inches below the soil surface?
- A. To minimize the escape of gaseous NH_3 into the air
 - B. To prevent it from reacting with soil water
 - C. To enhance soil acidity
 - D. To facilitate its faster absorption by plants

65. Which of the following reactions describes the hydrolysis of urea in soil?
- A. $(\text{NH}_2)_2\text{CO} + \text{H}_2\text{O} \rightarrow \text{CO}_2 + 2\text{NH}_3$ C. $\text{NH}_3 + \text{H}_2\text{O} \rightarrow \text{NH}_4\text{OH}$
 B. $(\text{NH}_4)_2\text{CO}_3 \rightarrow \text{CO}_2 + \text{NH}_4^+$ D. $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$
66. How does urea in UAN solutions react when incorporated into the soil?
- A. It volatilizes and escapes into the air
 B. It forms ammonia gas directly
 C. It decomposes into nitrogen and water
 D. It quickly reacts with soil water to form ammonium ions (NH_4^+)
67. What is a key characteristic of diammonium monohydrogen phosphate (DAP) as a fertilizer?
- A. It increases soil alkalinity C. It only provides phosphorus
 B. It temporarily increases soil acidity D. It is highly toxic to plants
68. Which is the correct sequence of steps involved in the industrial production of sulfuric acid using the contact process?
- A. Burning sulfur $\rightarrow$ Converting SO_3 to SO_2 $\rightarrow$ Absorption of SO_3 in H_2SO_4 $\rightarrow$ Dilution of oleum
 B. Converting SO_3 to SO_2 $\rightarrow$ Burning sulfur $\rightarrow$ Absorption of SO_3 in H_2SO_4 $\rightarrow$ Dilution of oleum
 C. Burning sulfur $\rightarrow$ Absorption of SO_3 in H_2SO_4 $\rightarrow$ Converting SO_2 to SO_3 $\rightarrow$ Dilution of oleum
 D. Burning sulfur $\rightarrow$ Converting SO_2 to SO_3 $\rightarrow$ Absorption of SO_3 in H_2SO_4 $\rightarrow$ Dilution of oleum
69. In the contact process, what is the catalyst used for the conversion of SO_2 to SO_3 ?
- A. Vanadium pentoxide (V_2O_5) C. Iron oxide (Fe_2O_3)
 B. Copper sulfate (CuSO_4) D. Platinum (Pt)
70. What is the primary characteristic of organophosphate pesticides?
- A. They are water-soluble and degrade quickly in the environment
 B. They are lipophilic and accumulate in fatty tissues
 C. They are highly toxic to vertebrates but less toxic to insects
 D. They are mainly used as household insecticides
71. Which of the following is true about neem leaf as a traditional pesticide?
- A. It is toxic to birds but safe for plants
 B. It is toxic to all types of insects
 C. It repels pests but is non-toxic to animals, birds, and humans
 D. It has a long residual activity and accumulates in the environment
72. What is the major difference between selective and non-selective herbicides?
- A. Selective herbicides kill all plant species, while non-selective herbicides kill only specific plants
 B. Selective herbicides target specific weeds, while non-selective herbicides kill all plants
 C. Selective herbicides are chemical, while non-selective herbicides are biological
 D. Selective herbicides affect animals, while non-selective herbicides affect only plants
73. What is the primary chemical composition of carbamate pesticides?
- A. Phosphorus compounds
 B. Nitrogen and carbon-based compounds
 C. Carbon, hydrogen, chlorine, and oxygen
 D. Compounds derived from plant alkaloids
74. What property of ceramics makes them useful for applications such as electrical insulators and dental replacements?
- A. High electrical conductivity C. Brittleness and chemical stability
 B. High magnetic properties D. High thermal expansion
75. Which of the following is a key material used to form traditional ceramics?
- A. Silicon carbide C. Barium titanate
 B. Clay D. Titanium carbide
76. What is the purpose of the densification process in ceramic manufacturing?
- A. To condense the ceramic into a strong, dense product
 B. To cool down the ceramic quickly
 C. To shape the ceramic into different molds
 D. To add impurities to the ceramic material
77. At what temperature range is a ceramic object typically heated during the densification process?
- A. 100-500°C C. 1000-1700°C
 B. 500-1000°C D. 1700-2000°C

78. What is the primary function of sodium carbonate in the glass production process?
- To increase the melting point of silica
 - To act as a flux in the melting of silica
 - To add color to the glass
 - To reduce the weight of the glass
79. Which type of glass is made using boron (III) oxide, B_2O_3 , instead of limestone or calcium oxide?
- Soda-lime glass
 - Borosilicate glass
 - Quartz glass
 - Lead glass
80. What type of glass would be produced if potassium carbonate was used instead of sodium carbonate?
- Soda-lime glass
 - Potassium glass
 - Lead glass
 - Borosilicate glass

CHEMISTRY ANSWER SHEET FOR GRADE 12 STUDENTS

Name _____ Sec. _____

I. Multiple choice

- | | | | |
|-----------|-----------|-----------|-----------|
| 1. _____ | 21. _____ | 41. _____ | 61. _____ |
| 2. _____ | 22. _____ | 42. _____ | 62. _____ |
| 3. _____ | 23. _____ | 43. _____ | 63. _____ |
| 4. _____ | 24. _____ | 44. _____ | 64. _____ |
| 5. _____ | 25. _____ | 45. _____ | 65. _____ |
| 6. _____ | 26. _____ | 46. _____ | 66. _____ |
| 7. _____ | 27. _____ | 47. _____ | 67. _____ |
| 8. _____ | 28. _____ | 48. _____ | 68. _____ |
| 9. _____ | 29. _____ | 49. _____ | 69. _____ |
| 10. _____ | 30. _____ | 50. _____ | 70. _____ |
| 11. _____ | 31. _____ | 51. _____ | 71. _____ |
| 12. _____ | 32. _____ | 52. _____ | 72. _____ |
| 13. _____ | 33. _____ | 53. _____ | 73. _____ |
| 14. _____ | 34. _____ | 54. _____ | 74. _____ |
| 15. _____ | 35. _____ | 55. _____ | 75. _____ |
| 16. _____ | 36. _____ | 56. _____ | 76. _____ |
| 17. _____ | 37. _____ | 57. _____ | 77. _____ |
| 18. _____ | 38. _____ | 58. _____ | 78. _____ |
| 19. _____ | 39. _____ | 59. _____ | 79. _____ |
| 20. _____ | 40. _____ | 60. _____ | 80. _____ |

